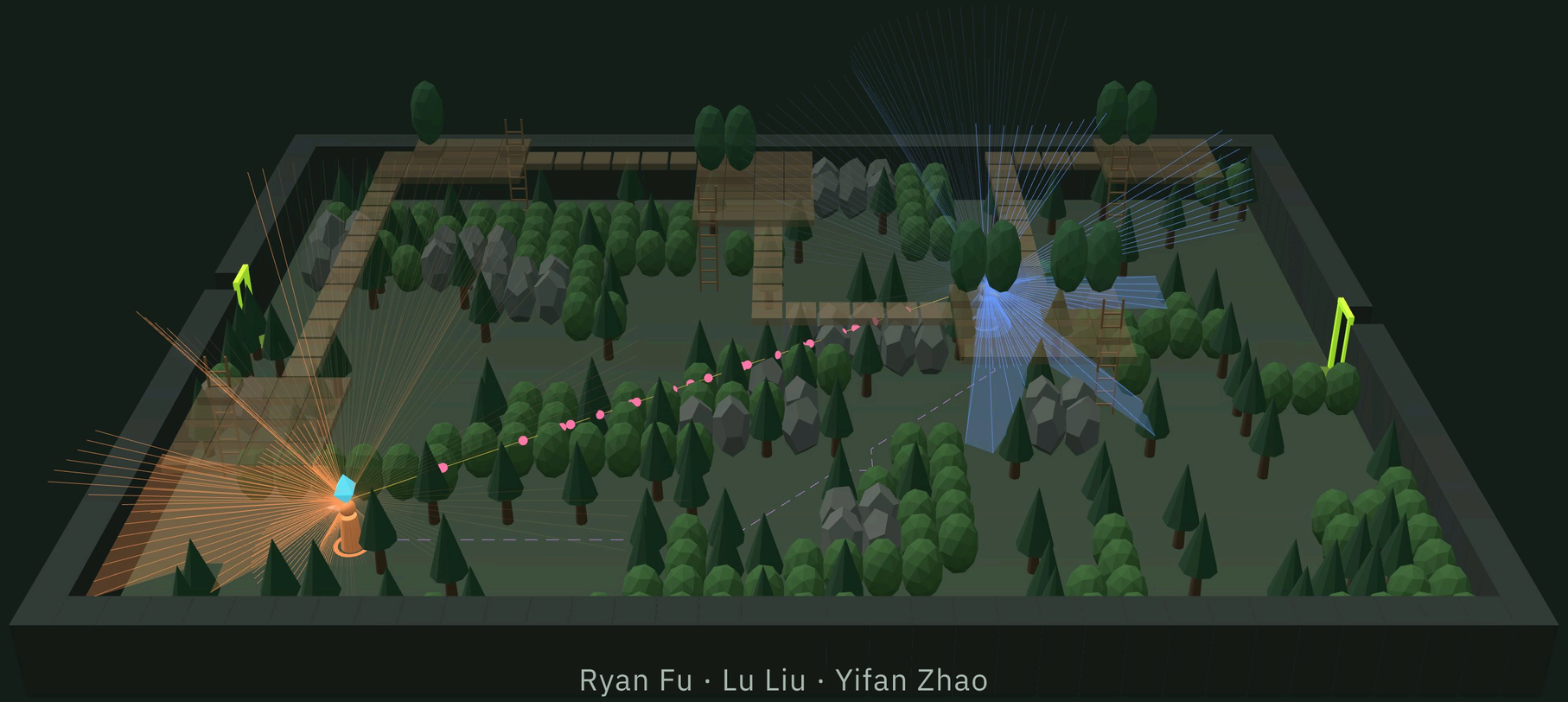
SPATIAL AI · COLUMBIA GSAPP · DISTANCE + PROXIMITY

Misty Forest Chase

A diamond heist where four distances drive four senses

Sight · Euclidean | Hearing · Acoustic | Walking · Path | Sense of place · Perceptual



Ryan Fu · Lu Liu · Yifan Zhao

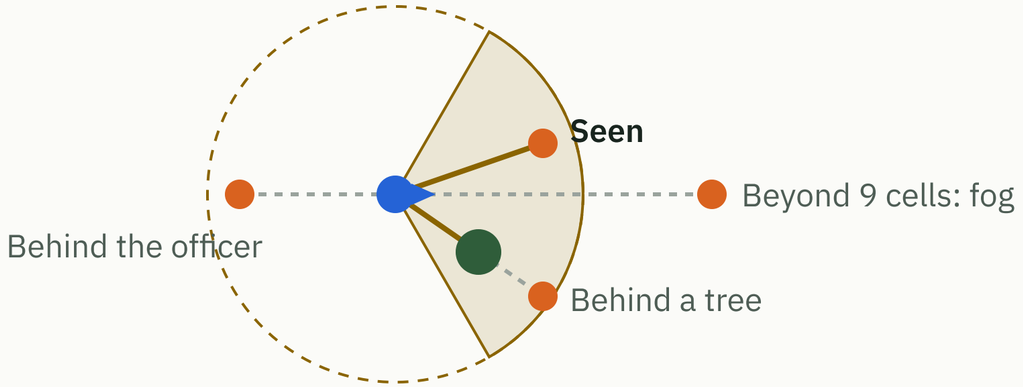
Seeing and hearing

SIGHT

Euclidean distance

$\sqrt{(\Delta x^2 + \Delta y^2 + \Delta z^2)}$

Seen only within 9 cells (the fog), inside a 120° forward cone, with nothing solid on the 3D line between the eyes.

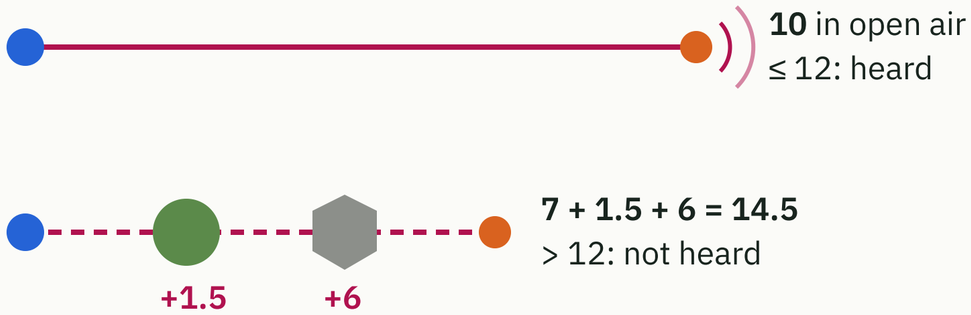


HEARING

Acoustic distance

`Euclidean + Σ muffling`

Heard within 12, in every direction. Each cell of material on the line adds: leaves 1, bush 1.5, tree 3, rock or floor 6.



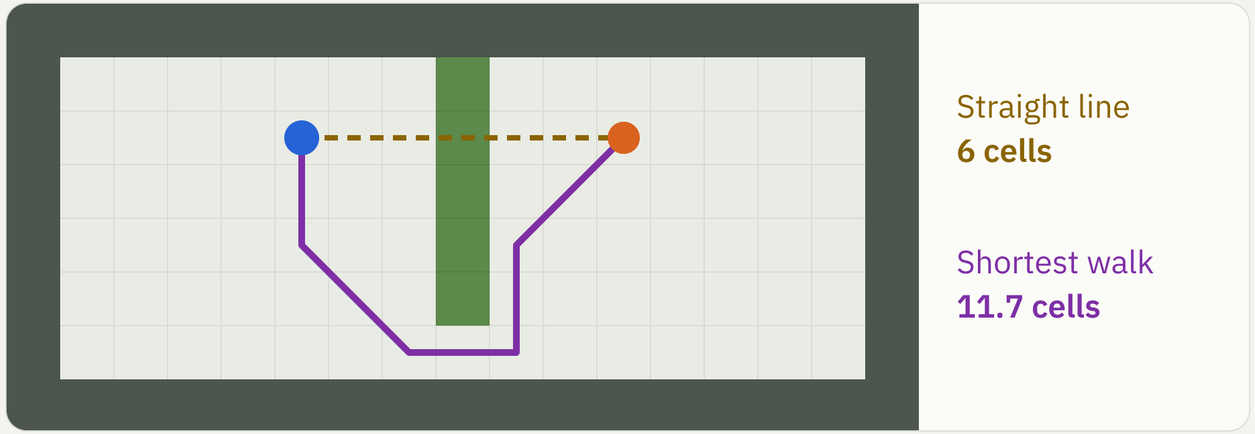
Walking and the sense of place

WALKING · THE BONUS

Path distance

`shortest walk (Dijkstra)`

Around obstacles and up ladders: 1 straight, √2 diagonal, 3 per ladder. Both AIs move along it.

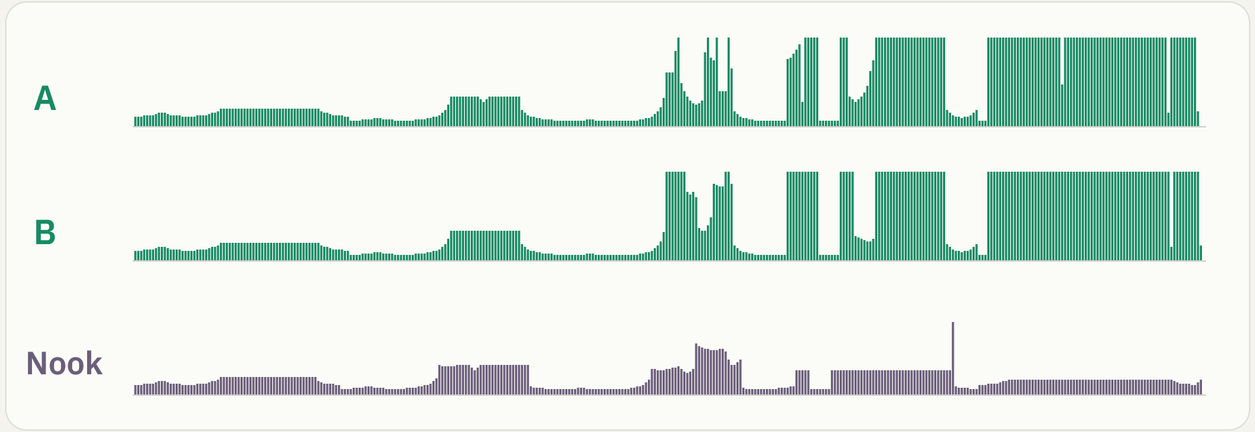


SENSE OF PLACE

Perceptual distance

`||r(a) - r(b)|| over 362 rays`

A and B are 15.5 cells apart, yet their fingerprints differ by only 22. A and a nook differ by 78.



What the distances showed

EUCLIDEAN

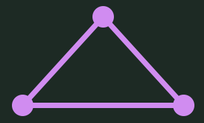
Metric



All four axioms hold

PATH

Metric



All four axioms hold

ACOUSTIC

Semimetric



Triangle fails in 598 of 4,000 tests

PERCEPTUAL

Pseudometric



Two places can look identical

49.7%

police wins over 320 rounds



- Caught before the theft · 39%
- Caught with the diamond · 10%
- Escaped after a chase · 20%
- Escaped without being seen · 30%